- *R family*—Memory optimized, with more memory than CPU power compared to the M family
- *X family*—Extensive capacity with a focus on memory, up to 1952 GB memory and 128 virtual cores
- *D family*—Storage optimized, offering huge HDD capacity
- *I family*—Storage optimized, offering huge SSD capacity
- *P, G, and CG family*—Accelerated computing based on GPUs (graphics processing units)
- *F family*—Accelerated computing based on FPGAs (field-programmable gate arrays)

Additional instance families are available for niche workloads like high-performance computing, in-memory databases, MacOS workloads, and more. See https://aws.amazon.com/ec2/instance-types/ for a full list of instance types and families.

Our experience indicates that you'll overestimate the resource requirements for your applications. We recommend that you try to start your application with a smaller instance type than you think you need at first—you can change the instance family and type later if needed.

Instance types and families

The names for different instance types are all structured in the same way. The instance family groups instance types with similar characteristics. AWS releases new instance types and families from time to time; the different versions are called generations. The instance size defines the capacity of CPU, memory, storage, and networking. For example, the instance type `t2.micro` tells you the following:

- The instance family is called `t`. It groups small, cheap virtual machines with low-baseline CPU performance but the ability to burst significantly over baseline CPU performance for a short time.
- You're using generation 2 of this instance family.